Aurick Anwar

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Education

McMaster University — Hamilton, Ontario
Bachelor of Engineering: Engineering Physics — *Expected 2029*

Work Experience

Machine Learning Intern — Toronto, Ontario
Personalized Prescribing — *Incoming Fall 2026*

- Building ML models to predict which medications will be effective based on **metabolic enzyme** and **brain receptor gene interactions.**

Software Engineering Intern — Toronto, Ontario
HermesAI — *May 2026 - Present*

- Built **React** interfaces for document upload and AI-powered case extraction, supporting legal files exceeding **1,000 pages**.
- Developed scalable legal document dashboards, reducing manual processing time by **50% for Canadian law firms.**
- Reduced document processing latency by **30%** by offloading file processing to asynchronous **Node.js.**

Founding Engineer — Toronto, Ontario
Magnified Systems — *February 2026 -Present*

- Built a motorcycle helmet-mounted **IMU (MPU6050)** device to detect and quantify impact severity in real time.
- Trained a **PyTorch** regression model on **1,000+ impact samples**, predicting severity within **±1%**.
- Developed a **C++** severity-score algorithm mapping linear and angular acceleration to a 1–100 impact scale.
- Designed the 3D modelled device enclosure in **SolidWorks**, producing a **PETG** fabricated model.

Personal Projects

Autonomous Self-Driving Vehicle with CARLA — *[CARLA, YOLOv11, PyTorch, OpenCV]*

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and **vehicle control,** achieving a real-time simulation at ~**20-30 FPS**.
- Wrote to a dataset with **5+ features** (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Used **CrossEntropyLoss**, achieving **~88.5%** prediction directional accuracy for precise automation navigation.

Breast Cancer Cell Detection — *[PyTorch, YOLOv11, Grad-Cam, CNNs]*

- Fine-tuned **YOLOv11** on **5,400+** annotated mammography images to localize breast cancer regions with bounding box detection across **benign** and **malignant** classes.
- Built a custom **3-layer PyTorch CNN** classifier from scratch achieving **95.31%** confidence on malignant tumor classifications using **CrossEntropyLoss** and **Adam optimization.**
- Implemented **Grad-CAM** explainability on the CNN's final convolutional layer, generating **spatial heatmaps** that visually highlight the tumor regions driving each prediction.

AI Smart Computer w/ Hand Gesture Control — *[MediaPipe, PyAutoGUI, OpenCV]*

- Implemented **6+ gesture mappings** with 21-point hand tracking, enabling click, cursor, mute, and volume control via distance thresholds.
- Reduced unintended inputs by **~60%** using gesture filtering and cooldown logic, achieving **<100 ms** latency.

Technical Skills

Programming Skills: Python, C++, React.JS, CSS, HTML, JavaScript, ROS2
Libraries: PyTorch, OpenCV, MediaPipe, YOLOv11, scikit-learn
Simulation: RVIZ, Gazebo, SLAM, Nvidia Isaac Sim
3D Design: AutoCAD, Fusion360, Inventor, SolidWorks
Electronics and Microcontrollers: Arduino, Raspberry PI, ESP32
Tools & OS: Docker, Linux, Ubuntu